Appendix 1: Alternative format for summarizing the evidence- Evidence Profiles

Evidence profiles, an alternative format for presenting summaries of evidence, contain the same key information as Summary of Findings Tables (SoFs). Evidence profiles have additional columns with decisions for each of the domains of certainty of evidence and may have a column specifying the importance of each outcome.

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	intensive antileukemic therapy	less intensive antileukemic therapy	Relative (95% CI)	Absolute (95% CI)		
Mortality (follow-up: range 7.7 months to 60 months; assessed with: Risk of death at any point in time)												
11	non-randomised studies	not serious[a]	not serious	not serious	not serious	none	1014 participants -	57.8%[b] 807 participants	HR 0.78 (0.69 to 0.89)	88 fewer per 1,000 (from 129 fewer to 42 fewer)	⊕⊕◯◯ Low	CRITICAL
Mortality (follow-up: 1 year; assessed with: Proportion of people who died)												
1	randomised trial	not serious	not serious	not serious	very serious[c]	none	22/44 (50.0%)	24/43 (55.8%) High risk	RR 0.83 (0.61 to 1.13)	95 fewer per 1,000 (from 218 fewer to 73 more)	⊕⊕◯◯ Low	CRITICAL
					serious[d]			20% Low risk		34 fewer per 1,000 (from 78 fewer to 26 more)	⊕⊕⊕◯ Moderate	
Mortality (follow-up: 1 year; assessed with: Proportion of people who died)												
11	non-randomised studies	not serious[a]	not serious	not serious	serious[d]	none	509/978 (52.0%)	391/676 (57.8%)	RR 0.93 (0.85 to 1.01)	40 fewer per 1,000 (from 87 fewer to 6 more)	⊕◯◯◯ Very low	CRITICAL
					not serious			20% Low risk		14 fewer per 1,000 (from 30 fewer to 2 more)	⊕⊕◯◯ Low	
Allogeneic hematopoietic stem cell transplantation (follow-up: range 8.5 months to 60 months; assessed with: Proportion of people who received transplantation)												
8	non-randomised studies	not serious[a]	not serious	not serious	not serious	very large effect[e]	172/747 (23.0%)	18/578 (3.1%)	RR 6.65 (4.13 to 10.71)	176 more per 1,000 (from 97 more to 302 more)	⊕⊕⊕◯ Moderate	IMPORTANT
Complete remission (follow-up: range 13.3 months to 60 months; assessed with: Duration in months (or time to relapse))												
4	non-randomised studies	not serious[a]	not serious	not serious	serious[f]	none	There were 4 studies reporting this outcome. Most of them (3) reported a shorter remission with IC. The difference in CR duration ranged from 3.1 months shorter to 0.03 months longer when patients received IC versus less-intensive therapy.				⊕◯◯◯ Very low	CRITICAL
Serious adverse events (follow-up: median 5 years; assessed with: Proportion of people who had Treatment emergent severe adverse events)												

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	intensive antileukemic therapy	less intensive antileukemic therapy	Relative (95% CI)	Absolute (95% CI)		
1	non-randomised studies	serious[g]	not serious	not serious	serious[h]	none	59/95 (62.1%)	44/95 (46.3%)	RR 1.34 (1.03 to 1.75)	157 more per 1,000 (from 14 more to 347 more)	⊕◯◯◯ Very low	IMPORTANT
Hospitalization (due to serious adverse events) (follow-up: range 20 months to 60 months; assessed with: Duration in days of hospitalization)												
3	non-randomised studies	not serious[a]	very serious[i]	not serious	very serious[j]	none	217	153	-	MD 12.96 days higher (16.23 lower to 42.15 higher)	⊕◯◯◯ Very low	IMPORTANT
Quality of life - not reported												
-	-	-	-	-	-	-	-	-	-	-	-	-

CI: confidence interval; **HR:** hazard ratio; **MD:** mean difference; **RR:** risk ratio

Explanations
a. Body of evidence from non-randomized studies, assessment of certainty of evidence started at low
b. Used event rate from 1-year mortality for the less-intensive therapy (from non-randomized studies).
c. Using a minimally important difference of 50 per 1000, the 95% CI suggests the possibility of an important benefit and an important harm.
d. Using a minimally important difference of 50 per 1000, the 95% CI suggests the possibility of a trivial effect and an important benefit.
e. The magnitude of effect is very large (point estimate 6.65)
f. A meta-analysis was not possible, but it is likely that the pooled estimate would have crossed the null effect.
g. The study was at serious risk of bias due to confounding and moderate risk of bias due to deviation from intended interventions so we rated down an additional level.
h. Using a minimally important difference of 10%, the 95% CI suggests the possibility of trivial and important harm
i. Not all confidence intervals of the studies overlap, and the statistical heterogeneity is high (I^2=99%)
j. Using a minimally important difference of 5 days, the 95% confidence interval suggests important benefit and important harm

Appendix 2. Writing footnotes to explain judgments of certainty of evidence

Explanatory footnotes should make the rationale for assessments of certainty of evidence transparent. They may also specify the source of the baseline risk used to calculate absolute effects. Explanatory footnotes should be brief, ideally no longer than two printed lines; provide appropriate information about reasons for judgments, additional information considered, and interpretation of results; created for the specific target audience; and be clear and self-explanatory.

The following examples provide suboptimal explanations, explain why these are not appropriate, and provide a better alternative:
For rating down for imprecision: *"Although the confidence interval does not cross the null, the confidence intervals were wide"*

Problems: does not clearly and explicitly justify rating down or not; not sufficiently explicit in that the term "wide" can mean different things depending on the thresholds set); and does not follow GRADE guidance for addressing imprecision.
A better explanation: "Although the 95% confidence interval does not cross the threshold of interest (null value), the point estimate suggests a large benefit and the body of evidence does not meet the optimal information size"

For rating down for inconsistency: "*I^2>30%, p value <0.1"*

Problems: This explanation is not informative (does not clearly and explicitly justify rating down, ">30%" can take many different values), relevant (many users may not be able to infer what this means), or accurate (does not seem to follow GRADE guidance for addressing inconsistency appropriately- As described in detail in Article 3 addressing inconsistency, GRADE users should never use the I^2 alone as the basis for rating down for inconsistency. Further, we do not recommend use of the p-value for the test of heterogeneity in Core GRADE (it adds little to the I^2 and is more difficult to interpret).
A better explanation: "Point estimates suggest large and important differences in effect, not all confidence intervals overlap, and I^2 is relatively high (59%). Subgroup analysis did not explain this inconsistency."

As a justification for not rating down for risk of bias: *"Three of the studies had high risk of bias due to inappropriate randomization, five had unclear risk of bias due to missing outcome data, three had high risk of bias in the outcome measurement, and one had high risk of bias due to selection of the reported result"*

Problems: this could be summarized appropriately without losing information; it is unclear why this led to rating down, as the authors have not specified what weight in the pooled estimate the three studies had and does not follow GRADE guidance for assessing risk of bias.
A better explanation: "Five of the studies, accounting for 32% of the weight of the pooled estimate, had high risk of bias in at least one domain, but their results were not importantly different than the results of the studies at low risk of bias and therefore we did not rate down".

Appendix 3. Calculating absolute effect based on baseline risk and relative effect

Here we show three examples of how one can calculate a risk difference and its associated confidence interval given a relative risk (example 1), an odds ratio (example 2), or a hazard ratio (example 3) and a control group risk. Article 1 in this series provides a glossary explaining the technical terms in this Appendix including absolute risk, relative risk, hazard ratio, risk difference.

Example 1

The meta-analyzed relative risk (RR) of 'disability worsening' in patients with multiple sclerosis from two RCTs (n = 1,479) comparing teriflunomide to placebo proved to be 0.76 (95% CI 0.62 – 0.93).[1]

The risk in the control group (estimated as the median risk in placebo groups) is 21%. When the number of included studies (denoted here as M) is even, the following formula can be used to calculate the median risk:

$$\frac{control\ group\ event\ rate\ in\ \frac{M}{2}th\ study\ +\ control\ group\ event\ rate\ in\left(\frac{M}{2}+1\right)th\ study}{2}$$

Where the studies, M, are sorted by increasing order of event rate.

In case of two control groups (M=2), the median risk is just the average of the first (here, 18.4%) and second risk (here, 23.5%) in the control groups. The median Risk = (18.4% + 23.5%)/2 = 21%.

(1) Risk with intervention (*corresponding risk)* = RR x Risk with control
= 0.76 x 21%
= 16%

(2) Risk difference = Risk with control – Risk with intervention
= 21% – 16%
= 5%

We can use the same procedure to calculate the confidence interval (CI) associated with the risk difference, substituting the extremes of the CI (here, 0.62 and 0.93) for the point estimate (here, 0.76).

(3) For the upper limit of the RR CI (0.93):
- Risk with intervention = 0.93 x 21% = 19.5%
- Risk difference = Risk with control – Risk with intervention = 21% – 19.5% = 1.5%

(4) For the lower limit of the RR CI (0.62):
- Risk with intervention = 0.62 x 21% = 13.0%
- Risk difference = Risk with control – Risk with intervention = 21% – 13.0% = 8.0%

(5) Risk difference is: 5% (95% CI: 1.5% to 8.0%).

Example 2

The odds ratio (OR) of 'incident hypertension' in pregnant women with overweight or obesity from four RCTs (n = 1,324) comparing exercise therapy + usual care to usual care is 0.52 (95% CI 0.28 – 0.96).[2]

The risk in the control group (estimated as the median risk in usual care groups) is 14%. When the number of included studies is even, the formula to calculate the median risk mentioned in example 1 can be applied. In case of four control groups (M=4), the median risk is just the average of the second (here, 9%) and third risk (here, 19.1%) in the control groups. The median risk = (9% + 19.1%)/2 = 14%.

To convert an odds ratio to a RR the next formula[3] can be used:

(1) $RR = \frac{OR}{(1-risk\ in\ control\ group) + (risk\ in\ control\ group \times OR)}$

$= \frac{0.52}{(1-0.14)+(0.14 \times 0.52)}$

= 0.56

We can use the same procedure to calculate to confidence interval (CI) associated with the odds ratio, substituting the extremes of the CI (here, 0.28 and 0.96) for the point estimate (here, 0.52).

(2) For the upper limit of the OR CI (0.96):

- RR = $\frac{0.96}{(1-0.14)+(0.14 \times 0.96)}$ = 0.97

(3) For the lower limit of the OR CI (0.28):

- RR = $\frac{0.28}{(1-0.14)+(0.14 \times 0.28)}$ = 0.31

Now we have the RRs of the point estimate (0.58) and its associated confidence interval (0.31 – 0.97) for a risk in the control group (estimated as the median risk in placebo groups) of 14%.

(4) Risk with intervention (*corresponding risk)* = RR x Risk with control
= 0.56 x 14%
= 7.8%

(5) Risk difference = Risk with control – Risk with intervention
= 14% – 7.8%
= 6.2%

We can use the same procedure to calculate the confidence interval (CI) associated with the risk difference, substituting the extremes of the CI (here, 0.31 and 0.97) for the point estimate (here, 0.56).

(6) For the upper limit of the RR CI (0.97):

- Risk with intervention = 0.97 x 14% = 13.6%
- Risk difference = Risk with control – Risk with intervention = 14% – 13.6% = 0.4%

(7) For the lower limit of the RR CI (0.31):

- Risk with intervention = 0.31 x 14% = 4.3%
- Risk difference = Risk with control – Risk with intervention = 14% – 4.3% = 9.7%

Risk difference is: 6.2% (95% CI: 0.4% to 9.7%).

Example 3

The hazard ratio (HR) of survival in patients with advanced hepatocellular carcinoma (n = 602) comparing sorafenib to placebo is 0.69 (95% CI 0.55 – 0.87).[4] Patients were followed up to about 17 months. We present two calculations. The first calculation relates to *event-free survival*. The second calculation relates to an *event*, i.e. death in this example.

The proportion of event-free patients up to about *eight* months in the control group is 0.50 (50%).
To convert a hazard ratio (HR) to a proportion of event-free patients in the intervention group the next formula[5] can be used:

(1) Proportion with intervention = $e^{\ln\,(control\ group\ event-free\ proportion)\ \times HR}$
= $e^{\ln(0.50)\ \times 0.69}$
= $e^{-0.6931\ \times 0.69}$
= $e^{(-0.4783)}$
= 0.6199 (i.e. 62.0% when rounded)

(2) Event-free survival difference = proportion event-free patients with intervention – proportion event-free patients with control
= 62% – 50% = 12.0% up to about eight months

We can use the same procedure to calculate the confidence interval (CI) associated with the hazard ratio, substituting the extremes of the CI (here, 0.55 and 0.87) for the point estimate (here, 0.69).

(3) For the upper limit of the CI (0.87):
- Proportion with intervention = $e^{\ln(0.50)\ \times 0.87}$ = 0.5471 (i.e. 54.7% rounded)
- Event-free survival difference = proportion event-free patients with intervention – proportion event-free patients with control
  = 54.7% – 50% = 4.7%

(4) For the lower limit of the CI (0.55):
- Proportion with intervention = $e^{\ln(0.50)\ \times 0.55}$ = 0.6830 (i.e. 68.3% rounded)
- Event-free survival difference = proportion event-free patients with intervention – proportion event-free patients with control
  = 68.3% – 50% = 18.3%

(5) Event-free survival difference up to about eight months is: 12.0% (95% CI: 4.7% to 18.3%).

The next section addresses calculating the occurrence of an event, i.e. death in this scenario.

As mentioned earlier, the second calculation relates to the occurrence of an event, i.e. death in this example; we use the same data as used for event-free survival. The proportion of patients with event in the control group [i.e. control group risk] up to about *ten* months is 0.625 (62.5%).

To convert a hazard ratio (HR) to an absolute risk in the intervention group the next formula[5] can be used:

(1) Risk with intervention = $1 - e^{\ln(1-control\ group\ proportion\ events) \times HR}$
= $1 - e^{\ln(1-0.625) \times 0.69}$
= $1 - 0.5083$ = 0.4917 (i.e. 49.2% when rounded)

(2) Risk difference = Risk with control – Risk with intervention
= 62.5% – 49.2%
= 13.3% up to about ten months

We can use the same procedure to calculate the confidence interval (CI) associated with the hazard ratio, substituting the extremes of the CI (here, 0.55 and 0.87) for the point estimate (here, 0.69).

(3) For the upper limit of the CI (0.87):

- Risk with intervention = $1 - e^{\ln(1-0.625) \times 0.87}$ = 0.5740 (i.e. 57.4% rounded)
- Risk difference = Risk with control – Risk with intervention = 62.5% – 57.4% = 5.1%

(4) For the lower limit of the CI (0.55):

- Risk with intervention = $1 - e^{\ln(1-0.625) \times 0.55}$ = 0.4169 (i.e. 41.7% rounded)
- Risk difference = Risk with control – Risk with intervention = 62.5% – 41.7% = 20.8%

(5) Risk difference up to about ten months is: 13.3% (95% CI: 5.1% to 20.8%).

References

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3. Prasad K, Jaeschke R, Wyer P, et al. Tips for teachers of evidence-based medicine: understanding odds ratios and their relationship to risk ratios. *J Gen Intern Med* 2008;23(5):635-40. doi: 10.1007/s11606-007-0453-4 [published Online First: 20080105]
4. Llovet JM, Ricci S, Mazzaferro V, et al. Sorafenib in advanced hepatocellular carcinoma. *N Engl J Med*. 2008;359(4):378-390. doi:10.1056/NEJMoa0708857
5. Tierney JF, Stewart LA, Ghersi D, Burdett S, Sydes MR. Practical methods for incorporating summary time-to-event data into meta-analysis. *Trials*. 2007;8:16. Published 2007 Jun 7. doi:10.1186/1745-6215-8-16

Appendix 4: Calculating and presenting absolute measures of effect: Directly calculating risk differences

There are scenarios in which deriving the risk difference from a relative effect and a baseline risk is problematic:

i) When the incidence of a binary outcome is low, it is possible to have a large relative effect despite a small or trivial absolute difference. Consider a study in which adverse events due to treatment occur in 1 in 1,000 patients in the comparator group and 3 in 1,000 in the treatment group. The relative risk of 3 appears large but the absolute effect of 2 in 1,000 more is extremely small.

Similarly, consider a situation in which deaths occur in 3 in 1,000 patients in the comparator group and in 1 in 1,000 patients in the intervention group. The relative risk is 0.33 representing a 67% or 2/3 relative risk reduction, a large relative effect. The absolute effect is however very small, only 2 in 1,000.

The result of this phenomenon is that one may see inflated estimates of absolute effect when one applies these large relative risks to external baseline risks that are higher than the incidence rates in the available studies (see Example 2 and to a lesser extent Example 1 below).

ii) The small number of events will typically yield very wide confidence intervals around the point estimate of relative risk. These confidence intervals are likely to be very asymmetrical. The reason is that one boundary of the confidence interval can go only from 1 to 0. The other boundary can be a very large number – 10, 50 or even over 100. The result may be a problematically asymmetric confidence interval with the side greater than 1 much larger than is appropriate (see all three examples below).

The most vivid example of the inappropriate asymmetry is when the point estimate is near a relative risk of 1 and one side of the confidence interval on the risk difference suggests a small benefit or harm and the other a very large benefit or harm (see example 3 below). Another consequence of the inappropriately wide confidence interval, rating down for imprecision twice instead of once and thus reducing certainty more than is warranted, occurs in all three examples.

iii) Trials may have 0 events in one or both arms. With 0 events in both arms the results are uninformative with respect to relative effects but very informative with respect to absolute effects: with 0 in both arms the risk difference is 0, the same risk difference one would see whenever the proportion of events is the same in intervention and control groups.

When there are 0 events in both groups whether to leave these trials out of the meta-analysis of relative effects or use a continuity correction remains controversial. Both approaches have limitations.

The solution to all these problems is to abandon our usual approach to estimation of absolute effects by applying baseline risk to relative effect estimates from a meta-analysis. Rather, GRADE users can conduct a meta-analysis of risk differences directly and use the results of this analysis to inform their pooled estimate of the risk difference and the associated 95% confidence interval.

In each example that follows we present first the usual approach applying the relative effects to the baseline risk and then the direct meta-analysis of risk differences. All three examples come from a World Health Organization guideline regarding the management of Arbovirus infections.[1]

Example 1

A systematic review considered the effects of platelet transfusion in patients with thrombocytopenia due to dengue virus addressed severe adverse events.[1] The authors identified two studies reporting on severe adverse events with a total of 460 patients and produced a relative risk of 4.63 (95% CI 0.78 to 27.4) from the meta-analysis. The baseline risk in the control group (no platelet transfusion) was 6 per 1000. Using the relative risk and baseline risk, the modeled risk difference was calculated as 22 more per 1000, with a 95% CI from 1 fewer to 158 more per 1000. Due to the wide confidence interval, the certainty of the evidence was rated down twice for imprecision.

Given the asymmetry of the modeled absolute effect's confidence interval, the authors conducted a meta-analyses of risk differences. This produced a risk difference of 15 more per 1000 with a 95% CI of 2 fewer to 32 more per 1000, leading to a more appropriate confidence interval requiring rating down only once for imprecision.

Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the Evidence (Quality of evidence)	Summary
		No platelet transfusion	Platelet transfusion		
Severe adverse events (applying relative effect to baseline risk)	Relative risk: 4.63 (CI 95% 0.78 - 27.4) Based on data from 460 participants in 2 studies Follow up 10.5 days	6 per 1000	28 per 1000	Very low Due to serious risk of bias[1] and to very serious imprecision.[2]	We are uncertain whether platelet transfusion increases severe adverse events
		Difference: 22 more per 1000 (CI 95% 1 fewer - 158 more)			
Severe adverse events (from meta-analysis of risk differences)	Relative risk: 4.63 (CI 95% 0.78 - 27.4) Based on data from 460 participants in 2 studies Follow up 10.5 days	6 per 1000	23 per 1000	Low Due to serious risk of bias[1] and serious imprecision.[3]	Platelet transfusions possibly have little to no increase in serious adverse events
		Difference: 15 more per 1000 (CI 95% 2 fewer - 32 more)			

[1] Risk of Bias: serious. Inadequate/lack of blinding of participants and personnel, resulting in potential for performance bias, Inadequate/lack of blinding of outcome assessors, resulting in potential for detection bias.
[2] Imprecision: very serious. 95% CI including absence and very large harms.
[3] Imprecision: serious. 95% CI including absence and modest harms.

Example 2

The same systematic review also assessed mortality as an outcome.[1] The meta-analysis, which included two studies and 460 patients, found a relative risk of 2.89 (95%CI 0.27 to 31.4). The baseline risk in the control group (no platelet transfusion) was 20 per 1000. Using this relative effect and baseline risk, the

authors calculated an absolute effect of 38 more deaths per 1000, with a 95% CI from 15 fewer to 608 more per 1000. The extremely wide confidence interval warranted a rating down of three levels for imprecision.

Considering the problematically asymmetrical confidence interval, the authors performed a meta-analysis of risk differences, yielding a risk difference of 11 more deaths per 1000, with a 95%CI from 13 fewer to 34 more per 1000. This resulted in a more symmetrical confidence interval but still required downgrading the certainty twice for imprecision.

Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the Evidence (Quality of evidence)	Summary
		No platelet transfusion	Platelet transfusion		
Mortality (applying relative effect to baseline risk)	Relative risk: 2.89 (CI 95% 0.27 - 31.4) Based on data from 460 participants in 2 studies	20 per 1000	58 per 1000	Very low Due to extremely serious imprecision.[1]	We are uncertain whether platelet transfusion increases or decreases mortality.
		Difference: 38 more per 1000 (CI 95% 15 fewer - 608 more)			
Mortality (from meta-analysis of risk differences)	Relative risk: 2.89 (CI 95% 0.27 - 31.4) Based on data from 460 participants in 2 studies	20 per 1000	31 per 1000	Low Due to very serious imprecision.[2]	Platelet transfusion possibly increases mortality.
		Difference: 11 more per 1000 (CI 95% 13 fewer - 34 more)			

[1] Imprecision: extremely serious, 95% CI including small benefits and very large harms. 2 events in total.
[2] Imprecision: very serious, 95% CI including modest benefits and harms. 2 events in total.

Example 3

The systematic review also considered organ failure as an outcome,[1] using data from one study with 372 patients. The meta-analysis reported a relative risk of 0.97 (95% CI 0.14 to 6.84). The baseline risk in the control group (no platelet transfusion) was 50 per 1000. Applying the relative effect and baseline risk, the calculated risk difference was 1 fewer per 1000, with 95%CI from 43 fewer to 292 more per 1000. This wide confidence interval led to a two-level downgrade for imprecision.

Given the asymmetrical confidence interval, the authors pooled the risk difference directly. The result was a risk difference of 1 fewer per 1000, with a 95%CI from 21 fewer to 21 more per 1000, requiring only one rating down for imprecision due to the more symmetrical confidence interval.

Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the Evidence (Quality of evidence)	Summary
		No platelet transfusion	Platelet transfusion		
Organ failure (applying relative effect to baseline risk)	Relative risk: 0.97 (CI 95% 0.14 - 6.84) Based on data from 372 participants in 1 study	50 per 1000	49 per 1000	Very low Due to serious risk of bias[1] and to very serious imprecision.[2]	We are uncertain whether platelet transfusion increases or decreases organ failure.
		Difference: 1 fewer per 1000 (CI 95% 43 fewer - 292 more)			

	Follow up 21 days				
Organ failure (from meta-analysis of risk differences)	Relative risk: 0.97 (CI 95% 0.14 - 6.84) Based on data from 372 participants in 1 study Follow up 21 days	50 per 1000	49 per 1000	Low Due to serious risk of bias[1] and serious imprecision.[3]	Platelet transfusion possibly has little or no impact on organ failure.
		Difference: 1 fewer per 1000 (CI 95% 21 fewer - 21 more)			

[1] Risk of Bias: serious. Inadequate/lack of blinding of participants and personnel, resulting in potential for performance bias, Inadequate/lack of blinding of outcome assessors, resulting in potential for detection bias.
[2] Imprecision: very serious. 95% CI small and large harms. 11 events.
[3] Imprecision: serious, 95% CI including modest benefits and harms. 11 events.

Reference

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Appendix 5. Practical issues presentations including in SoF

The burden of treatment and other practical issues related to how a treatment or test affects the patients' daily life are frequently missing in guidelines and in many tools for shared decision-making.
These practical issues have also been called the "work" of being a patient. An exploration of large database of patient's reports of burden of treatment and practical issues related to interventions, has identified 15 categories of practical issues [1]:

1. **Medication routine** (e.g., frequency and route)
2. **Tests and visits** (i.e., how many are required to implement an intervention or as follow-up)
3. **Procedure and device** (i.e., what procedures are required, are there any devices involved)
4. **Recovery and adaptation** (i.e., how will recovery typically look like, what additional effort is required, how to manage dependencies)
5. **Coordination of care** (i.e., how does an intervention practically fit with the rest of care)
6. **Adverse effect, drug interactions and antidotes** (i.e., are there specific adverse effects or interactions to manage or anticipate, what is involved to mitigate or manage them, are there existing antidotes to drug interventions)
7. **Physical well-being** (i.e., is there any physical distress that may come with managing practical issues relation to an intervention)
8. **Emotional well-being** (i.e., is there any emotional distress that may come with managing practical issues relation to an intervention)
9. **Pregnancy and nursing** (i.e., how may an intervention be adjusted to pregnancy and nursing)
10. **Costs and access** (i.e., who is paying, and what is required to access the recommendation, assuming it is available in one's context)
11. **Food and drinks** (e.g., are there specific directions regarding diet or other health habits)
12. **Exercise and activities** (e.g., how may an intervention impact one's ability to function in daily life or to exercise)
13. **Social life and relationships** (e.g., how may an intervention impact one's social life and interactions)
14. **Work and education** (e.g., how may an intervention impact one's work or education)
15. **Travel and driving** (e.g., how may an intervention impact the ability to travel or drive a vehicle)

Having data on the categories, which are relevant for an intervention, summarized in SoFs can serve support their translation into **tools for shared decision making with patients.** [2 3]

Examples of how this could be done are displayed in the next page:

A. Example of a SoF comparing SGLT-2 inhibitors vs Standard of Care, including numerical summary of the evidence of each outcome, followed by a display of relevant practical issue

(Source : MAGICapp: https://app.magicapp.org/#/guideline/5288)

Sodium-Glucose Transport Protein 2 (SGLT2) inhibitors vs Standard care

Adults with type 2 diabetes plus 3 more risk factors and no underlying CVD or CKD

10 Outcomes **11** Practical issues

Outcome Timeframe	Study results and measurements	Absolute effect estimates Standard care	 SGLT2 inhibitors	Certainty of the evidence (Quality of evidence)	Plain language summary
All-cause mortality	Odds ratio 0.77 (CI 95% 0.71 — 0.83) Based on data from 282704 participants in 225 studies	70 per 1000	55 per 1000 Difference: **15 fewer per 1000** (CI 95% 19 fewer — 11 fewer)	High	SGLT2 inhibitors reduce the risk of death compared with standard care.
Cardiovascular mortality	Odds ratio 0.84 (CI 95% 0.76 — 0.92) Based on data from 222944 participants in 128 studies	46 per 1000	39 per 1000 Difference: **7 fewer per 1000** (CI 95% 11 fewer — 4 fewer)	Moderate Due to serious imprecision.	SGLT2 inhibitors probably reduce the risk of cardiovascular death compared with standard care.
Nonfatal myocardial infarction	Odds ratio 0.87 (CI 95% 0.79 — 0.97) Based on data from 262168 participants in 199 studies	58 per 1000	51 per 1000 Difference: **7 fewer per 1000** (CI 95% 12 fewer — 2 fewer)	Moderate Due to serious imprecision	SGLT2 inhibitors probably reduce the risk of nonfatal myocardial infarction compared with standard care.

Sodium-Glucose Transport Protein 2 (SGLT2) inhibitors vs Standard care

Adults with type 2 diabetes plus 3 more risk factors and no underlying CVD or CKD

10 Outcomes **11** Practical issues

Practical issues	SGLT2 inhibitors	Standard care	Both
Medication routine	Tablets swallowed once daily at the same time. Some need to be taken in the morning.	Most anti-diabetic medications are tablets except insulin, an injection.	
Tests and visits	Needs to monitor blood pressure and kidney function.	Needs self-monitoring of blood glucose at home if the patients are receiving insulin, sulfonylureas or meglitinides.	Regular blood samples, and more frequently after starting a new medication. We only consider adding these drugs when the patient has HbA1C above 6.5.
Recovery and adaptation			Most people need to take diabetes medications for the rest of their lives. Should not be taken while a person is sick, especially if there is vomiting, diarrhea, or the person isn't eating and drinking very much.
Adverse effects, interactions and antidote	Mild to moderate urogenital infections can occur. Low blood pressure, dizziness, dehydration, etc., especially among people older than 65 years or when combined with diuretics. May not be as effective in patients with very poor kidney function. All patients should monitor for signs of diabetic ketoacidosis (even if the blood glucose is within the normal range) and seek hospital attention immediately if it occurs.	Adverse effects vary between specific agents, especially for insulin or insulin secretagogues can cause low blood sugar.	SGLT2 inhibitors or GLP-1 RAs do not increase the risk of hypoglycaemia when taken alone. But they do increase the risk of hypoglycemia when used in combination with some medications such as insulin or insulin secretagogues.

B. Example of how key practical issue practical issues can be displayed in an infographic

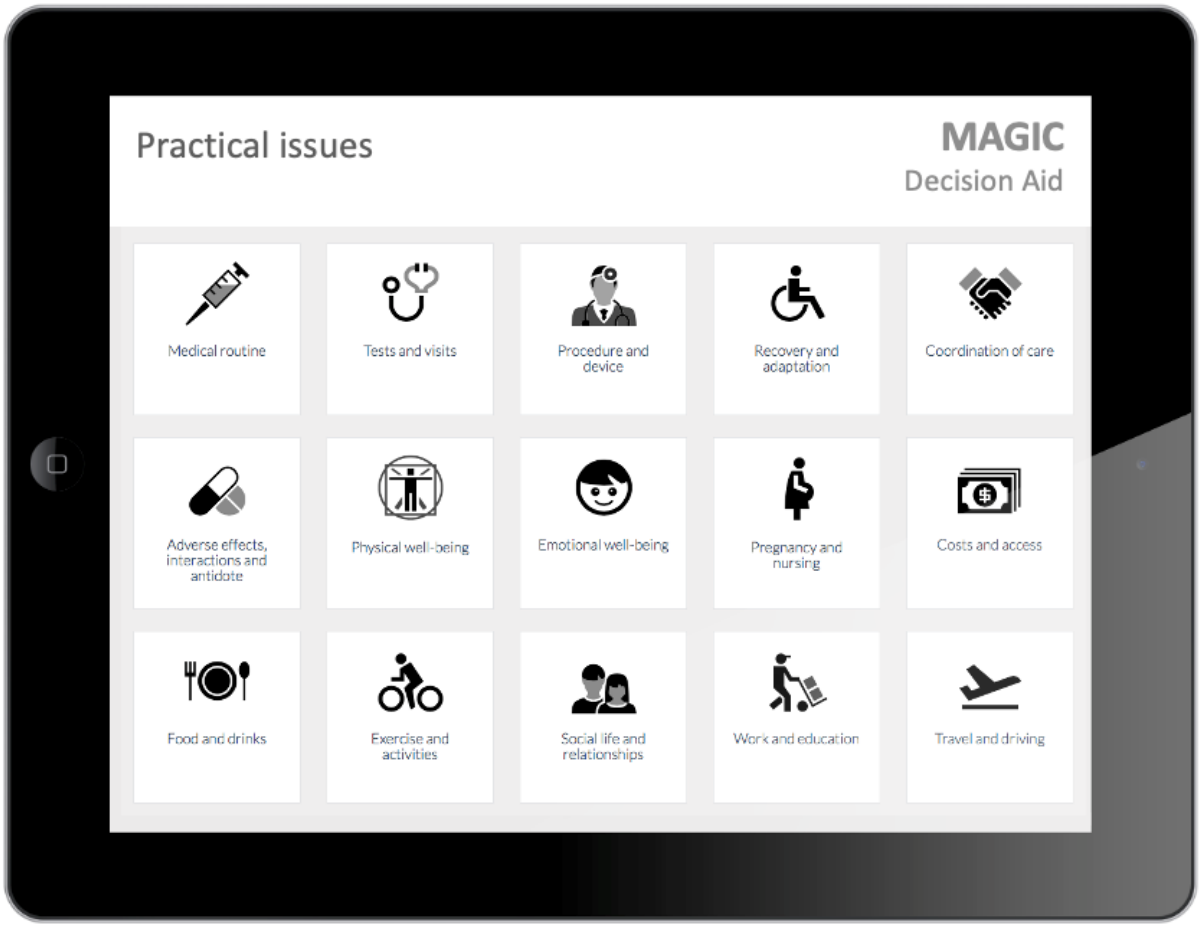
(Source : BMJ Rapid Recommandation : https://www.bmj.com/content/373/bmj.n1091)

Key practical issues

Usual care	SGLT-2 inhibitors	GLP-1 receptor agonists
Most anti-diabetic medications are tablets except insulin, an injection	Should not be taken while a person is unwell, especially if there is vomiting, diarrhoea, or the person isn't eating and drinking much	
	Tablets swallowed once daily. Some must be taken in the morning	Injection once or twice daily or once weekly
		Also available as tablets, but these are not yet widely available
		Formulas that combine GLP-1 receptor agonists and insulin are available

C. Example of practical issue displayed in a Decision Aid to support shared decision making in the clinical encounter

(Source: SHARE-IT Decision aids in MAGICapp: https://magicevidence.org/decision-aids)



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